Digital Signal Processing **HK261** **CO2036**

DIGITAL SIGNAL PROCESSING LAB REPORT

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Overview

This report summarizes the results of the Digital Signal Processing laboratory exercises, specifically focusing on generating, manipulating, and analyzing discrete-time signals and systems using Scilab. The report covers Lab 1 and Lab 2.

1 Introduction

The laboratory exercises provide practical experience with continuous-time to discrete-time signal conversion, signal properties, and elementary discrete-time operations. By utilizing Scilab, theoretical concepts such as sampling, quantization, time-shifting, folding, and signal arithmetic are modeled and visualized to deepen our mathematical understanding.

2 Objectives

The objectives achieved upon completing these laboratory sessions include:

- Utilizing Scilab for vector and matrix initializations.
- Understanding the sampling theorem and the quantization process (ADC procedure).
- Visualizing discrete signals via functions such as `plot2d3` and utilizing graph annotations.
- Implementing fundamental operations on discrete-time signals, including addition, multiplication, time-shifting, and time-reversal.

- Extracting the even and odd components of a given discrete-time sequence.

3 Methodology

The exercises were conducted on a Windows environment using the Scilab numerical computing software. Mathematical formulas from the time-domain representation of discrete signals were translated into Scilab scripts. Vectorized operations and padding techniques were applied where necessary to align discrete sequences prior to manipulation.

4 Practical Contents

4.1 Lab 1

Exercise 1.1

Question: Using the operators in vector and matrix to do the following tasks:

- Create a vector in the form of $(x_1 + 1, x_2 + 1, x_3 + 1, x_4 + 1)$ where x_1, x_2, x_3, x_4 are components of vector $x = 1 : 4$.
- Create a vector in the form of $(x_1y_1, x_2y_2, x_3y_3, x_4y_4)$ where $x_1 \dots x_4$ and $y_1 \dots y_4$ are components of vector $x = 1 : 4$ and vector $y = 5 : 8$, respectively.
- Create a vector in the form of $(\sin(x_1), \sin(x_2), \dots, \sin(x_{10}))$ where x is a vector of 10 values linearly chosen in the interval $[0, \pi]$.

Explanation: This exercise introduces element-wise vector operations in Scilab.

- For the first task, adding a scalar to a vector adds that scalar to every element identically: $x + 1 \Rightarrow [2, 3, 4, 5]$.
- The second task requires element-by-element multiplication (the Hadamard product). In Scilab, this is executed using the `.*` operator, multiplying elements at corresponding indices: $x. * y \Rightarrow [1 \times 5, 2 \times 6, 3 \times 7, 4 \times 8]$.
- The third task utilizes the `linspace(0, %pi, 10)` function to generate an array of $N = 10$ linearly spaced points. The `sin()` function is then mapped across the generated vector space.

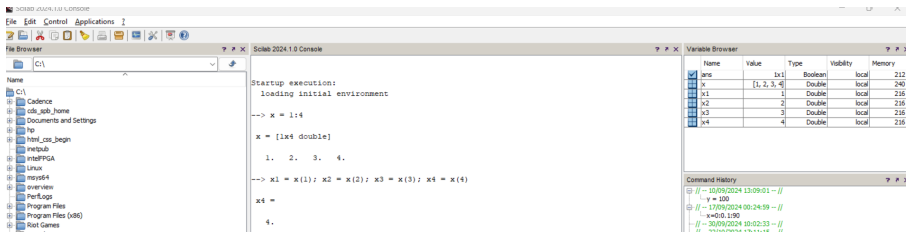


Figure 1: Result of Ex 1.1 - Part 1

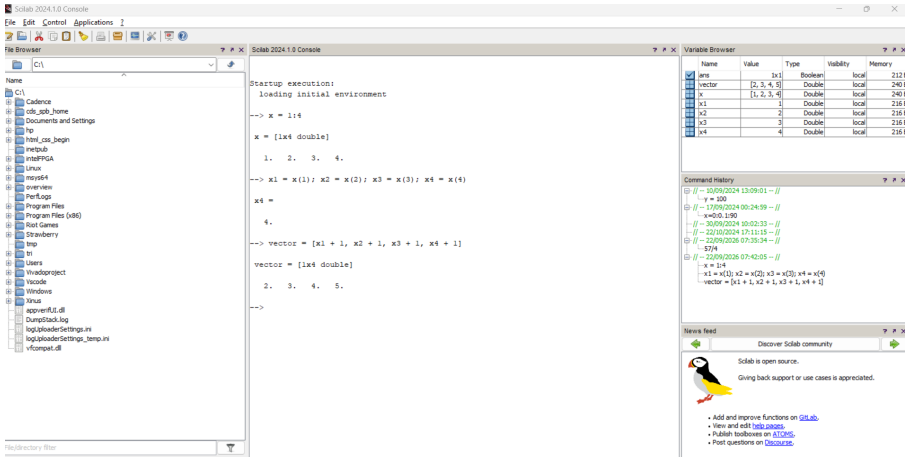


Figure 2: Result of Ex 1.1 - Part 2

Exercise 1.2

Question: Consider the analog signal $x_a(t) = 3\sin(100\pi t)$.

- Use Scilab to draw $x_a(t)$ in 5 periods.
- Determine the discrete-time signal $x(n)$ of the signal $x_a(t)$ that is sampled with a sampling rate $F_s = 300$ samples/s.
- Determine the discrete-time signal $x(n)$ and determine its periodic property. If $x(n)$ is a periodic signal, determine the frequency and period of $x(n)$. Then, use Scilab to draw $x(n)$ in 5 periods.
- Determine the quantized signal $x_q(n)$ if $\Delta = 0.1$ using the truncated method. Then, draw $x_q(n)$ in 5 periods. All figures must be displayed in a single window.

Explanation: Given $x_a(t) = 3\sin(100\pi t)$, the analog frequency is $F_0 = 50$ Hz. By applying the sampling theorem with $F_s = 300$ Hz, the sampling interval is $T_s = 1/F_s$. Substituting $t = nT_s$:

$$x(n) = x_a(nT_s) = 3\sin\left(100\pi\frac{n}{300}\right) = 3\sin\left(\frac{\pi}{3}n\right) \tag{1}$$

A discrete sinusoid is periodic if its digital frequency $f = \frac{F_0}{F_s} = \frac{50}{300} = \frac{1}{6}$ is a rational number. Because $f = \frac{1}{6}$, $x(n)$ is periodic. The fundamental period is determined by $N = \frac{K}{f}$ where K is the smallest integer yielding an integer N. Thus, for $K = 1$, the discrete period is $N = 6$ samples.

For quantization, we apply a truncation method with step size $\Delta = 0.1$. The truncated value $x_q(n)$ is obtained via the floor operation after normalizing by Δ:

$$x_q(n) = \Delta\lfloor\frac{x(n)}{\Delta}\rfloor = 0.1\lfloor 10 \cdot x(n)\rfloor \tag{2}$$

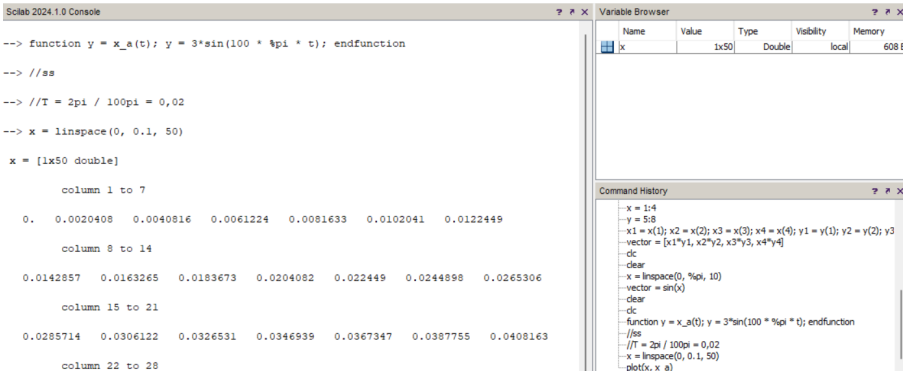


Figure 3: Result of Ex 1.2 - Analog and Discrete Signal

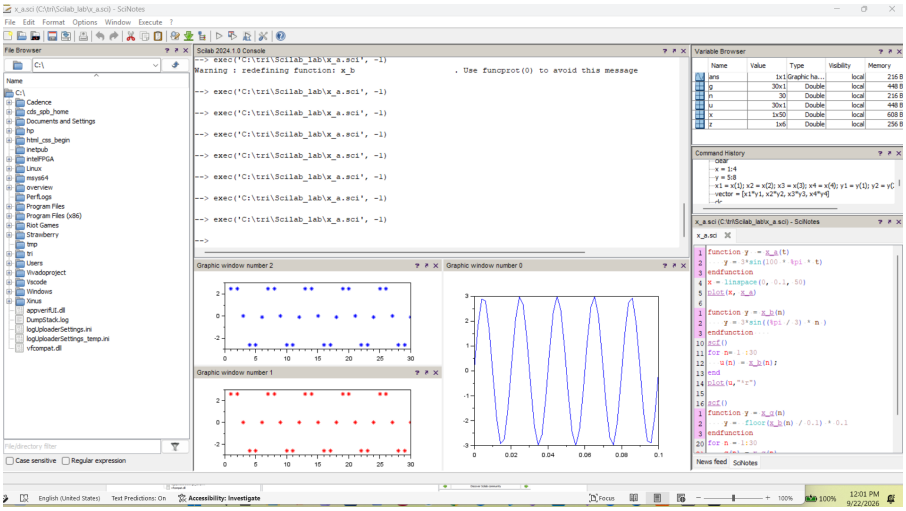


Figure 4: Result of Ex 1.2 - Quantized Signal and Errors

4.2 Lab 2

Exercise 1: Investigate Basic Functions

Question: Let's investigate the following functions on Scilab and briefly report their functionalities and how to use it: `plot2d3`, `min`, `max`, `subplot`, `title`, `xlabel`, `ylabel`, `bool2s`, `deff`.

Explanation: This exercise evaluates fundamental plotting and computational functions:

- `plot2d3(x, y)`: Plots vertical lines (stems) representing discrete-time samples. It is strictly used for discrete signals to highlight values at integer indices.
- `max()` and `min()`: Calculates the maximum/minimum elements in a matrix or vector array, highly useful for determining normalization constants or quantization step limits.
- `subplot(m, n, p)`: Divides the graphic window into an $m \times n$ matrix of sub-windows, selecting the pth window for the active plot.

- `title()`, `xlabel()`, `ylabel()`: These functions add textual annotations to the graph. `title()` sets the main heading, while `xlabel()` and `ylabel()` label the respective axes, which is crucial for identifying discrete components properly.
- `bool2s()`: Converts a boolean array containing %T and %F into a numerical array of 1s and 0s. This is mathematically equivalent to the Heaviside step function evaluating a logic condition.
- `deff()`: Defines inline custom functions on the fly, similar to lambda expressions. Syntax: `deff("[y] = f(x)", "y = ...")`.

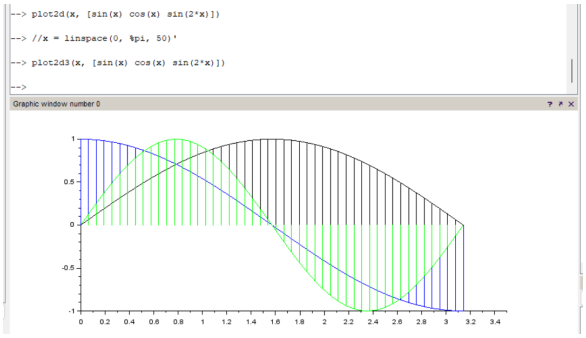


Figure 5: plot2d3 Usage

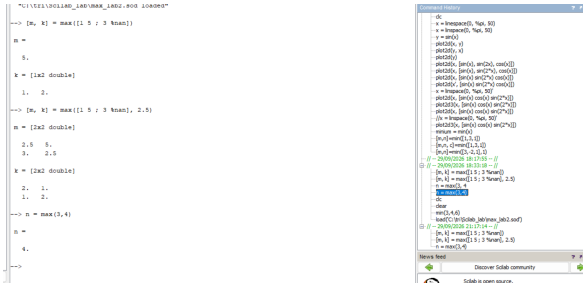


Figure 6: max() Usage

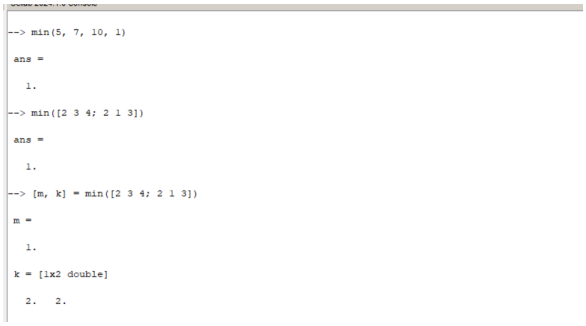


Figure 7: min() Usage

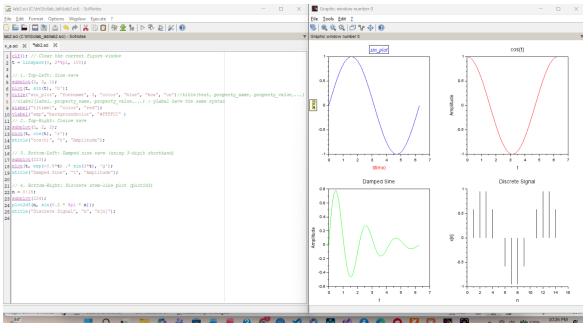


Figure 8: subplot() + title() + xlabel() + ylabel() Usage

```
--> deff("y = f(x)", " y = 3*x + 1");

--> f(5)

 ans =

   16.
```

Figure 9: deff Usage

Exercise 2 & Exercise 3

Question: Try the following scripts on Scilab and report your understanding after observing the output.

- `n = -5:5; msignal = bool2s(n >= 0); plot2d3(n, msignal)`
- `n = -5:5; msignal = bool2s(n == 0); plot2d3(n, msignal)`

Explanation:

- **Exercise 2:** The logical condition `n >= 0` evaluates to True for $n \geq 0$ and False otherwise. `bool2s` transforms these booleans into numeric 1s and 0s. The output precisely replicates the mathematical definition of a **unit step sequence** $u(n)$:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{3}$$

- **Exercise 3:** The equality `n == 0` is true only at the origin point $n = 0$. Consequently, applying `bool2s` creates a sequence that represents the **unit sample sequence (discrete Dirac delta function)** $\delta(n)$:

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \tag{4}$$

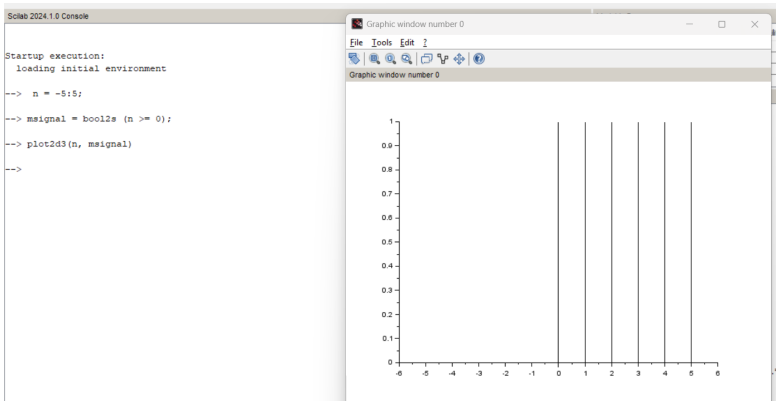


Figure 10: Result of Ex 2 - Unit Step Sequence $u(n)$

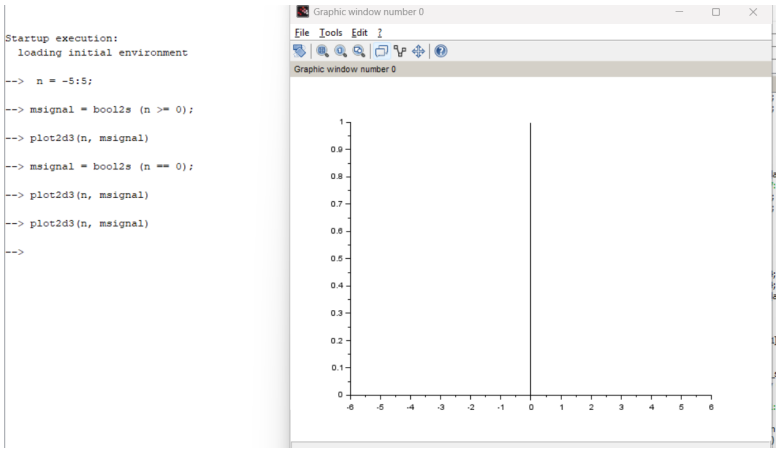


Figure 11: Result of Ex 3 - Unit Sample Sequence $\delta(n)$

Exercise 4: Unit Ramp Signal

Question: Use Scilab to draw the unit ramp signal $u_r(n)$ for $n = -5 : 5$.

Explanation: The discrete unit ramp signal is mathematically defined as the integral of the unit step function, or more precisely:

$$u_r(n) = n \cdot u(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{5}$$

In Scilab, this is implemented natively by executing an element-wise product between the index space array n and the previously defined unit step array.

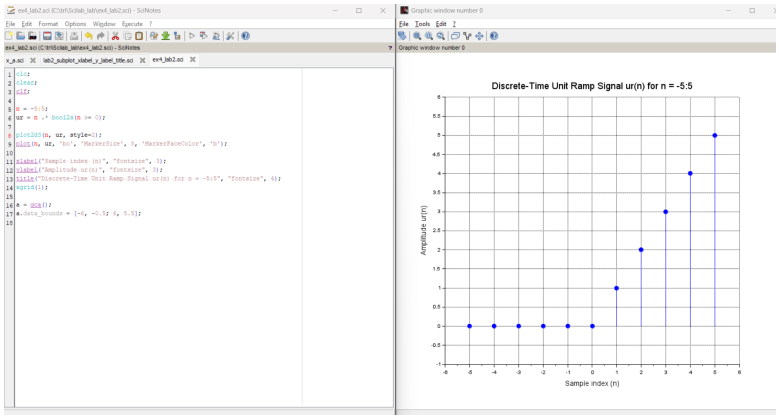


Figure 12: Result of Ex 4 - Unit Ramp Signal

Exercise 5: Even and Odd Components

Question: Given a discrete-time signal $x(n) = \{1, 3\uparrow, -2\}$. Use Scilab to draw the signal $x(n)$, the odd signal component $x_o(n)$, and the even signal component $x_e(n)$.

Explanation: The arrow signifies the origin index $n = 0$. The sequence is thus explicitly indexed as $x(-1) = 1, x(0) = 3, x(1) = -2$. According to signal properties, any arbitrary sequence can be perfectly decomposed into an even-symmetric component $x_e(n)$ and an odd-symmetric component $x_o(n)$. We first evaluate the folded (time-reversed) sequence $x(-n)$. Using the decomposition formula:

$$x_e(n) = \frac{x(n) + x(-n)}{2}, \quad x_o(n) = \frac{x(n) - x(-n)}{2} \tag{6}$$

Calculations yield $x_e(-1) = -0.5, x_e(0) = 3, x_e(1) = -0.5$, and correspondingly $x_o(-1) = 1.5, x_o(0) = 0, x_o(1) = -1.5$. Scilab strictly mirrors these mathematical operations linearly. Therefore, the resulting sequences are:

$$x_e(n) = \{-0.5, 3\uparrow, -0.5\}, \quad x_o(n) = \{1.5, 0\uparrow, -1.5\} \tag{7}$$

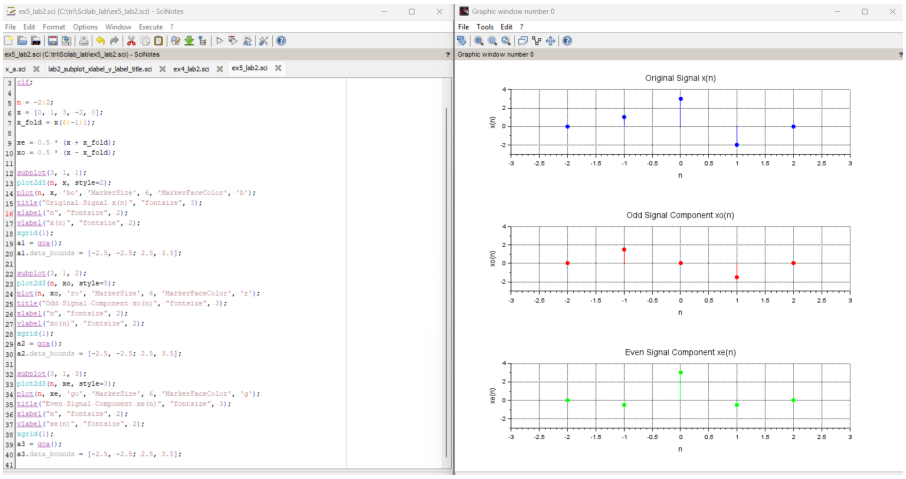


Figure 13: Result of Ex 5 - Even and Odd Decomposition

Exercise 6 & 7: Addition and Multiplication of Signals

Question:

- **Exercise 6:** Given two discrete-time signals $x_1(n) = \{0\uparrow, 1, 3, -2\}$ and $x_2(n) = \{0, 1\uparrow, 2, 3\}$. Determine $y(n) = x_1(n) + x_2(n)$.
- **Exercise 7:** Given the same signals, determine $y(n) = x_1(n) \cdot x_2(n)$.

Explanation: Because the two sequences span different bounded domain intervals ($x_1(n)$ spans $[0, 3]$ and $x_2(n)$ spans $[-1, 2]$), they cannot be directly operated on using primitive arrays. To implement a rigorous sample-by-sample operation, we determine the union of both index domains: $[\min(n_{1_{min}}, n_{2_{min}}), \max(n_{1_{max}}, n_{2_{max}})] = [-1, 3]$. Both vectors are then symmetrically zero-padded to conform to this unified domain grid. Once the sequences structurally align relative to the time origin $n = 0$:

$$y_{add}(n) = x_1(n) + x_2(n) = \{0, 1\uparrow, 3, 6, -2\} \tag{8}$$

$$y_{mult}(n) = x_1(n) \cdot x_2(n) = \{0, 0\uparrow, 2, 9, 0\} \text{ or simply } \{0\uparrow, 2, 9\} \tag{9}$$

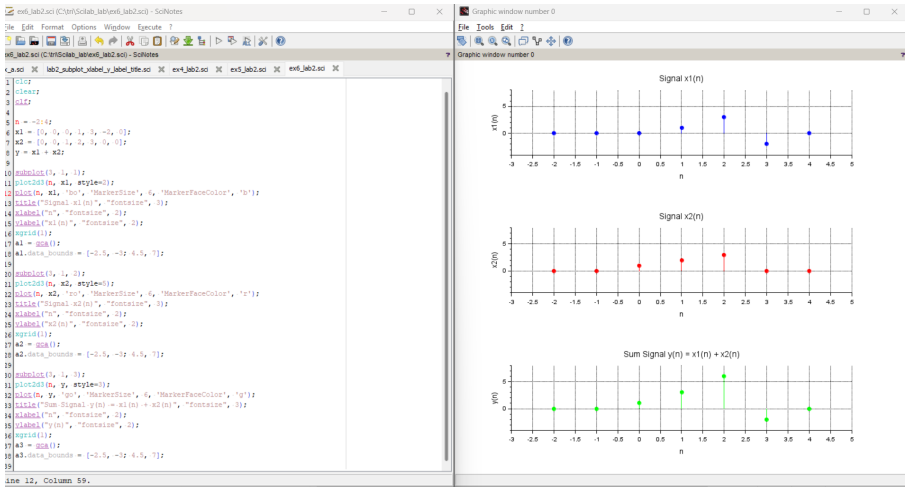


Figure 14: Result of Ex 6 - Addition of Signals

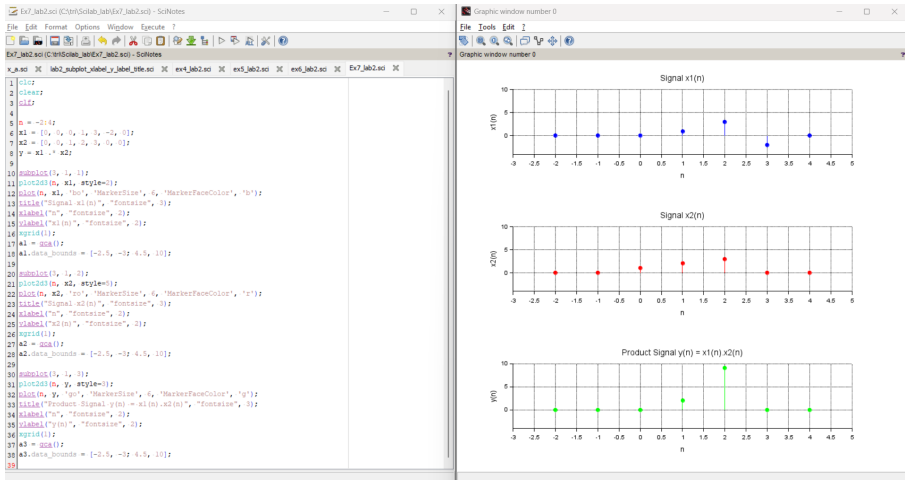


Figure 15: Result of Ex 7 - Multiplication of Signals

Exercise 8: Signal Manipulations

Question: Given a discrete-time signal $x(n) = \{1, -2, 3\uparrow, 6\}$. Determine the following signals and draw both original and manipulated signals: 1. $y_1(n) = x(-n)$ 2. $y_2(n) = x(n+3)$ 3. $y_3(n) = 2x(-n-2)$

Explanation: For $x(n)$, the origin translates to $x(-2) = 1, x(-1) = -2, x(0) = 3, x(1) = 6$. We systematically apply independent operations on the sequence index variables:

1. **Folding (Time-Reversal):** $y_1(n) = x(-n)$. Every index evaluates to its symmetrical negative counterpart. Resulting indices: $[-1, 2]$.

$$y_1(n) = \{6, 3\uparrow, -2, 1\} \tag{10}$$

2. **Advance (Time-Shift):** $y_2(n) = x(n+3)$. The addition inside the argument denotes an advancement. The whole sequence is shifted to the left along the time axis by 3 sample units.

$$y_2(n) = \{1, -2, 3, 6, 0, 0\uparrow\} \tag{11}$$

3. **Composite Operations:** $y_3(n) = 2x(-(n+2))$. To evaluate this correctly: first perform folding to get $x(-n)$, then shift left by 2 to get $x(-(n+2))$, and finally apply an amplitude scaling factor of 2.

$$y_3(n) = \{12, 6, -4, 2\uparrow\} \tag{12}$$

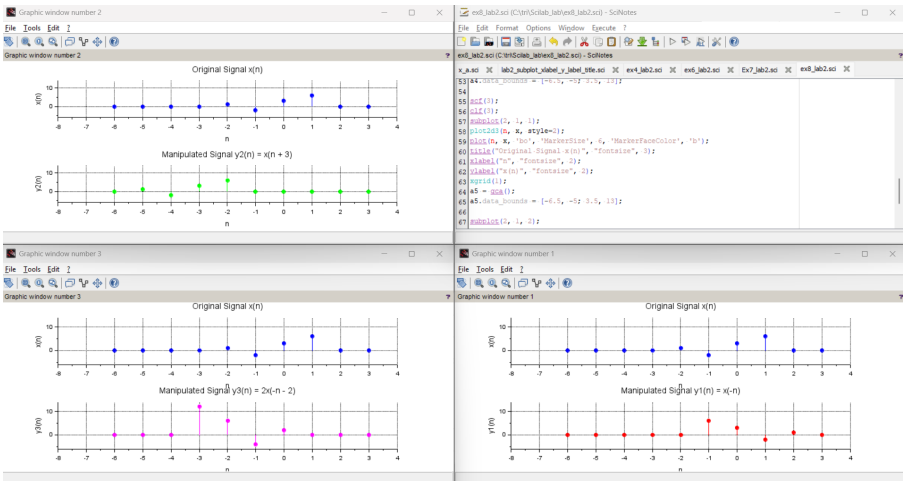


Figure 16: Result of Ex 8 - Signal Manipulations

Additional exercise 1.1

Classify the following signals according to whether they are (1) one- or multi-dimensional; (2) single or multichannel, (3) continuous time or discrete time, and (4) analog or digital (in amplitude). Give a brief explanation.

(a) Closing prices of utility stocks on the New York Stock Exchange.

(b) A color movie.

(c) Position of the steering wheel of a car in motion relative to car's reference frame.

(d) Position of the steering wheel of a car in motion relative to ground reference frame.

(e) Weight and height measurements of a child taken every month.

Answer:

(a) (1) One dimensional because closing prices are functions of time.
(2) Multichannel because the signal is a vector of prices.
(3) Discrete-time because closing prices are defined only at a specific instance of time.
(4) Digital because prices are money, which is discrete-valued.

(b) (1) Multidimensional because light characteristics at each point is a function of 2-D coordinates and time.
(2) "Movie" in this context refers to traditional film, which is single channel.
(3) Continuous-time because light characteristics are defined at every point in time as the film rolls.
(4) Analog because light has a continuous physical properties.

(c) (1) One dimensional because position is a function of time.
(2) "Position" in this context refers to the angular rotation of the steering wheel, which means single channel.
(3) Continuous-time because position is defined at every point in time.
(4) Analog because angular position is a continuous physical quantity.

(d) (1) One dimensional, (2) single channel, (3) continuous-time, (4) analog.

(e) (1) One dimensional because height and weight are functions of time.
(2) Multichannel because the signal consists of height and weight.
(3) Discrete-time because height and weight are only measured monthly.
(4) Digital because measured height and weight are discrete valued while physical height and weight are analog valued.

Additional exercise 1.3

Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x_a(t) = 3\cos(5t + \pi/6)$

(b) $x(n) = 3\cos(5n + \pi/6)$

(c) $x(n) = 2\exp[j(n/6 - \pi)]$

(d) $x(n) = \cos(n/8)\cos(\pi n/8)$

(e) $x(n) = \cos(\pi n/2) - \sin(\pi n/8) + 3\cos(\pi n/4 + \pi/3)$

Answer:

(a) $T_p = \frac{1}{F} = \frac{2\pi}{5}$. We have

$$\begin{aligned} x_a(t+T_p) &= 3cos(2\pi\frac{5}{2\pi}t + 2\pi + \frac{\pi}{6}) \\ &= 3cos(5t + \frac{\pi}{6}) \\ &= x_a(t) \\ &\Rightarrow Periodic \end{aligned}$$

(b) $x(n) = 3\cos(5n + \pi/6) = 3\cos(2\pi\frac{5}{2\pi}n + \pi/6)$. $f = \frac{5}{2\pi}$ is not rational, therefore not periodic.

(c) $x(n) = 2\exp[j(n/6 - \pi)] = 2e^{j(2\pi\frac{1}{12\pi}n - \pi)}$. $f = \frac{1}{12\pi}$ is not rational, therefore not periodic.

(d) $\cos(n/8)$ is nonperiodic, $\cos(\pi n/8)$ is periodic, their product is nonperiodic.

(e) $x(n) = \cos(\pi n/2) - \sin(\pi n/8) + 3\cos(\pi n/4 + \pi/3) = \cos(2\pi\frac{1}{4}n) - \sin(2\pi\frac{1}{16}n) + 3\cos(2\pi\frac{1}{8}n + \pi/3)$. $\cos(2\pi\frac{1}{4}n)$ is periodic with $N = 4$, $\sin(2\pi\frac{1}{16}n)$ is periodic $N = 16$, $3\cos(2\pi\frac{1}{8}n + \pi/3)$ is periodic with $N = 8$. Therefore $x(n)$ is periodic with $N = LCM(4, 16, 8) = 16$.

Additional exercise 1.6

A continuous-time sinusoid $x_a(t)$ with fundamental period $T_p = 1/F_0$ is sampled at a rate $F_s = 1/T$ to produce a discrete-time sinusoid $x(n) = x_a(nT)$.

(a) Show that $x(n)$ is periodic if $T/T_p = k/N$ (i.e., T/T_p is a rational number).

(b) If $x(n)$ is periodic, what is its fundamental period T_p in seconds?

(c) Explain the statement: $x(n)$ is periodic if its fundamental period T_p, in seconds, is equal to an integer number of periods of $x_a(t)$.

Answer:

(a) We have

$$\frac{T}{T_p} = \frac{1/F_s}{1/F_0} = \frac{F_0}{F_s}$$

and

$$x(n) = A\cos(2\pi\frac{F_0}{F_s}n + \theta)$$

Thus, $x(n)$ is periodic if T/T_p is rational.

(b) When $x(n)$ is periodic, we have $f_0 = F_0/F_s = k/N \rightarrow N = k/f_0$. Suppose the fundamental period of $x(n)$ is $T_d = NT$, we have $T_d = (\frac{k}{f_0})T = k(\frac{T_p}{T})T = kT_p$. Thus it takes kT_p seconds to make 1 period T_d of $x(n)$.

(c) $T_d = kT_p \Leftrightarrow NT = kT_p \Rightarrow f = k/N = T/T_p \Rightarrow$ f is rational $\Rightarrow x(n)$ is periodic.

Additional exercise 1.7

An analog signal contains frequencies up to 10 kHz.

(a) What range of sampling frequencies allows exact reconstruction of this signal from its samples?

(b) Suppose that we sample this signal with a sampling frequency $F_s = 8$ kHz. Examine what happens to the frequency $F_1 = 5$ kHz.

(c) Repeat part (b) for a frequency $F_2 = 9$ kHz.

Answer:

(a) $F_s \geq 2F_{max} = 20\,kHz$.

(b) With $F_s = 8\,kHz$, only signals in the range $-F_s/2 \leq F \leq F_s/2$ can be reconstructed exactly. The frequency $F_1 = 5\,kHz$ is folded with the folding frequency $F_s/2 = 4\,kHz$ $F_1' = F_1 - kF_s = 5000 - k8000 = -3000\,Hz$ (with $k = 1$).

(c) $F_2 = 9\,kHz$ gets folded to $F_2' = F_2 - kF_s = 9000 - k8000 = 1000\,Hz$ (with $k = 1$).

Additional exercise 1.9

An analog signal $x_a(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per second.

(a) Determine the Nyquist sampling rate for $x_a(t)$.

(b) Determine the folding frequency.

(c) What are the frequencies, in radians, in the resulting discrete time signal $x(n)$?

(d) If $x(n)$ is passed through an ideal D/A converter, what is the reconstructed signal $y_a(t)$?

Answer:

(a) $F_N = 2B = 2 \cdot 360 = 720\,Hz$.

(b) $F_s/2 = 600/2 = 300\,Hz$.

(c) We have

$$x(n) = \sin(\frac{480\pi}{600}n) + 3\sin(\frac{720\pi}{600}n) = \sin(\frac{4\pi}{5}n) + 3\sin(\frac{6\pi}{5}n)$$

thus $f_1 = \frac{2}{5}\,Hz$, $f_2 = \frac{3}{5}\,Hz$, $\omega_1 = \frac{4\pi}{5}$, $\omega_2 = \frac{6\pi}{5}$.

(d) We have

$$\begin{aligned} x(n) &= \sin(\frac{4\pi}{5}n) + 3\sin(\frac{6\pi}{5}n) \\ &= \sin(\frac{4\pi}{5}n) + 3\sin(-\frac{4\pi}{5}n + 2\pi n) \\ &= \sin(\frac{4\pi}{5}n) - 3\sin(\frac{4\pi}{5}n) \\ &= -2\sin(\frac{4\pi}{5}) \end{aligned}$$

then $y_a(t) = x(F_s t) = -2\sin(480\pi t)$.

Additional exercise 1.10

A digital communication link carries binary-coded words representing samples of an input signal

$$x_a(t) = 3\cos 600\pi t + 2\cos 1800\pi t$$

The link is operated at 10,000 bits/s and each input sample is quantized into 1024 different voltage levels.

(a) What are the sampling frequency and the folding frequency?

(b) What is the Nyquist rate for the signal $x_a(t)$?

(c) What are the frequencies in the resulting discrete-time signal $x(n)$?

(d) What is the resolution Δ?

Answer:

(a)

$$\begin{aligned}\text{Number of bits/}sample &= \log_2(1024) = 10\\ F_s &= \frac{10000\text{ bit/second}}{10\text{ bit/sample}}\\ &= 1000\text{ sample/second}\\ F_{fold} &= F_s/2 = 500\ Hz\end{aligned}$$

(b) $F_N = 2B = 2 \cdot 900 = 1800\ Hz$.

(c) $F_1 = 300\ Hz < 500\ Hz$ so it does not get folded. $F_2 = 900\ Hz > 500\ Hz$ so it gets folded to $F_2' = -100\ Hz$. We have

$$\begin{aligned}x(n) &= 3\cos(2\pi\frac{300}{1000}n) + 2\cos(2\pi\frac{-100}{1000}n)\\ &= 3\cos(2\pi\frac{3}{10}n) + 2\cos(2\pi\frac{1}{10}n)\end{aligned}$$

thus $f_1 = 3/10\ Hz$, $f_2 = 1/10\ Hz$.

(d) $\Delta = (x_{max} - x_{min})/(L-1) = (5-(-5))/1023 = 10/1023$.

5 Discussion

The laboratory thoroughly consolidated the theoretical understanding of discrete-time signals using systematic computational approaches. Scilab proved to be a versatile platform, allowing direct vector mapping of discrete mathematical formulas and offering visual validations through its `plot2d3` subroutines. Specifically, manual implementations of zero-padding demonstrated exactly how sequences must be mutually aligned relative to an origin reference prior to algebraic operations.

6 References

- Digital Signal Processing Lecture Notes - Ho Chi Minh City University of Technology.
- Scilab Online Manuals.

7 Source Code

The source code, scripts, and LaTeX files for this laboratory report can be accessed at our GitHub repository:
https://github.com/aulequangtri/signal_scilab